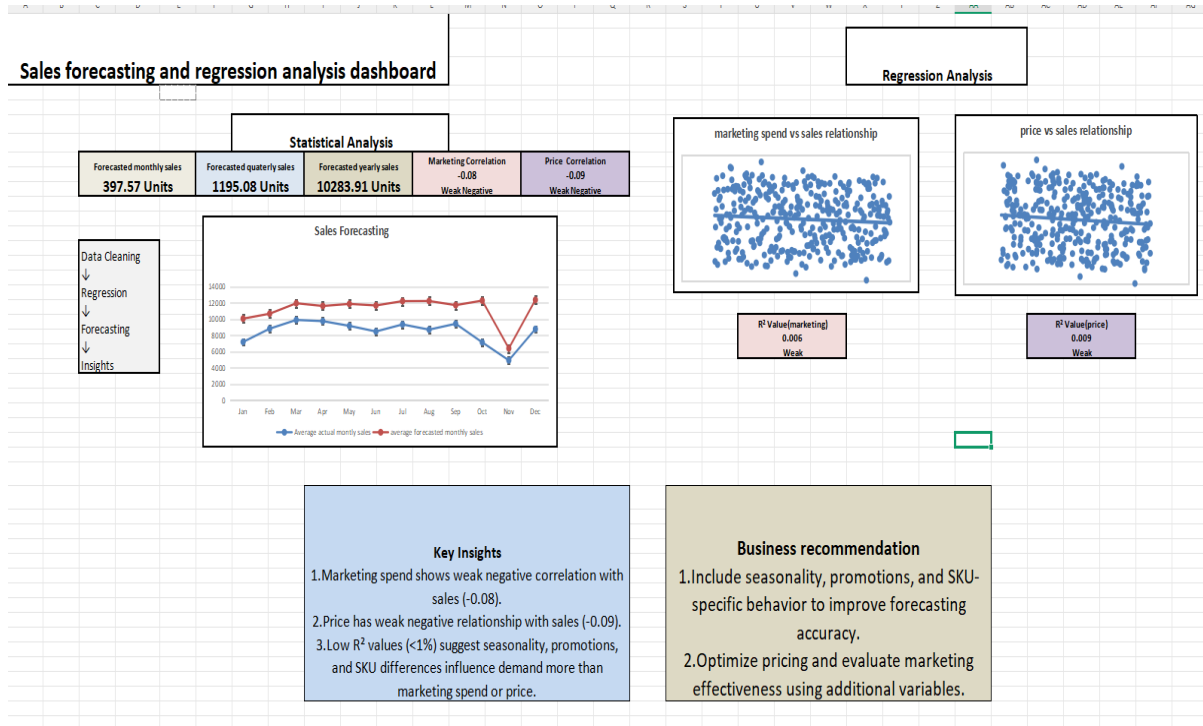


Sales Forecasting & Regression Analysis Report



Objective

The objective of this project was to analyze historical sales data and forecast future demand using **regression analysis and moving averages**. The study examined the impact of **Marketing Spend** and **Price** on **Sales** and predicted future sales for the next month, quarter, and year.

Methodology

- Cleaned data by removing outliers using **Box & Whisker plots**
- Performed **multiple regression analysis** with Marketing Spend and Price as predictors of Sales
- Used **3-month moving averages** to forecast future Marketing Spend and Price
- Calculated **Correlation** and **R² values** to measure relationships between variables
- Created a dashboard with forecasts, trend graphs, regression analysis, and insights

Key Findings

- Marketing Spend and Sales showed a **weak negative correlation (-0.08)**
- Price and Sales also showed a **weak negative correlation (-0.09)**
- Very low **R² values (0.006 and 0.009)** indicate that Marketing Spend and Price alone explain little variation in sales
- Other factors such as **seasonality, promotions, and SKU behavior** likely influence demand more strongly

Forecast Results

Next Month Sales: 397.57 units

Next Quarter Sales: 1195.08 units

Next Year Sales: 10283.91 units

Recommendation

Future forecasting models should include seasonality, promotions, and SKU-level trends to improve prediction accuracy and support better business decisions.

Conclusion

This project demonstrates the application of **data cleaning, regression analysis, forecasting, and dashboard visualization** to support demand forecasting and data-driven decision making.